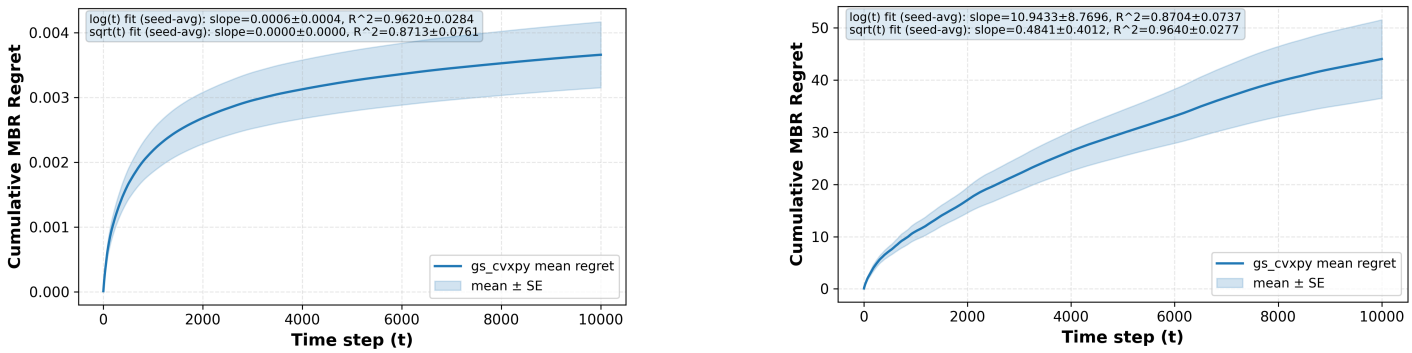
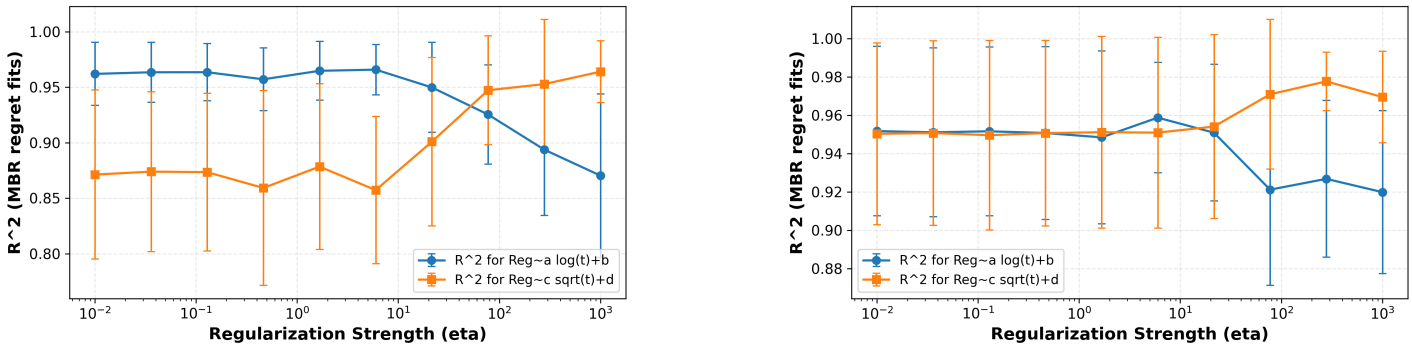
Empirical Validation of Greedy Sampling (GS)



(a) Small $\eta = 0.01$: $\log T$ regime

(b) Large $\eta = 1000$: $\sqrt{T}$ regime

Figure 1: **MBR Regret Trajectories.** In a GBPM setting ($K = 20$, $T = 10^4$, $d = 5$, $r = 1$, reverse-KL regularizer), regret tightly fits a $\log T$ curve for small η and shifts to a $\sqrt{T}$ curve for large η, corroborating Theorem 4.2.



(a) $d = 5$

(b) $d = 10$

Figure 2: **Crossover Point Analysis.** Goodness-of-fit (R^2) comparison. Theorem 4.2 dictates the $\sqrt{T}$ regime dominates when $\eta \gtrsim d^{-2}\sqrt{T}$. Increasing dimension from $d = 5$ to 10 empirically shifts this crossover to a smaller η.

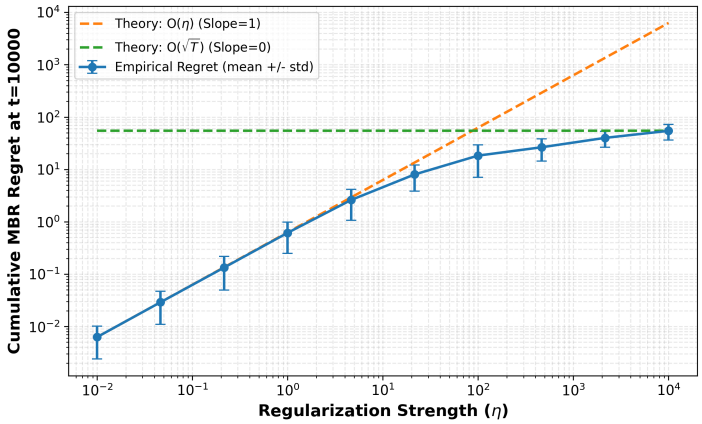


Figure 3: **Final Regret Scaling.** Cumulative MBR regret at $T = 10^4$ vs. η ($d = 10$, $r = 1$). Regret initially grows linearly but strictly plateaus for large η, perfectly matching the unified $\min(\cdot, \cdot)$ bound.